Predicting Life Expectancy

Machine Learning Driven Assessment of Global Life Expectancy Trends

WHO Life Expectancy Dataset · 193 Countries · 2000 – 2015



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1-The Learning Task

Motivation:

We believe that life expectancy is far more than just a statistical figure; it is a fundamental reflection of the quality of human life and the environment we inhabit. Driven by our shared passion for longevity and the dream of a healthier, longer-living humanity, we chose to tackle this challenge by moving beyond traditional demographic models. By applying machine learning to real-world WHO and UN data, we are not only capturing the complex, non-linear relationships that influence human health but also identifying the key factors that truly matter, with the ultimate goal of translating data into a more prosperous future for everyone.



Task Definition:

Type: Supervised Learning

Target: Life Expectancy (Years)

Data: 2900 Country-Year Records

2-Data Analysis

Dataset Scope: Analyzed 193 countries across 16 years (2000–2015), resulting in ~2,900 high-quality, cleaned records.

Methodology: Processed raw data by handling missing values through country-specific mean and median imputation. Standardized features to ensure model stability across all three ML approaches.

Enhanced Feature Set: Beyond the core indicators, we engineered predictive variables, including *Dev_Index* (Schooling × Income) and *Immunity_Index*, to capture deeper socioeconomic and health dynamics.

Key Drivers:

Positive Correlation: Education and income levels are the primary engines of longevity (+0.75 and +0.72).

Negative Correlation: High adult mortality and disease prevalence (HIV/AIDS) remain the most significant barriers to life expectancy (-0.70 and -0.59).

3 · Literature Overview

Dolgopolyi et al. (2024)

Interpretable ML for Life Expectancy

Linear Regression R^2=0.73 · Decision Tree R^2=0.90 · Random Forest R^2=0.94. Key drivers: diphtheria, measles, adult mortality.

Lakshmanarao et al. (2022)

Immunization & HDI Factors via ML

Tested LR, SVM, Decision Tree, Random Forest. RF reached R^2=0.96. Immunization features (Polio, Diphtheria) ranked among top predictors.

Roslan et al. (2024)

Socio-Health Factors via ML

Schooling correlation r=0.752, the strongest positive predictor found. Confirms HIV/AIDS & adult mortality as top negative drivers.

Gulkhayo Urinova (Linear Regression)

Load & Clean → Impute Missing → Encode + Scale → 70/30 Split → Train Model → Evaluate

Baseline: Linear Regression

A straight-line model: it fits one weight per feature and adds them up to estimate life expectancy. Simple, fast, and easy to explain — my starting point.

R^2 = 0.8195 · MAE = 2.98 years

My Model: LASSO Regression

Same idea as Linear Regression, but with a built-in filter: LASSO shrinks weak features' weights toward zero, automatically selecting which inputs really matter.

α (optimal) = 0.0016 · 5-fold CV

Why this matters: features like the two "thinness" columns are nearly identical (r = 0.94) — LASSO is built to detect and shrink this kind of redundancy automatically.

5 · Results & Analysis

Model	R^2	MAE (yrs)	RMSE (yrs)	Features Kept
Linear Regression (OLS)	0.8195	2.98	4.11	16 / 16
LASSO Regression	0.8195	2.98	4.11	16 / 16 (0 zeroed)

Both models tie

OLS and LASSO score identically — LASSO confirms my original feature set was already well-built, rather than improving on it.

Top drivers confirmed

HIV/AIDS, Adult Mortality (negative) and Schooling, Income composition (positive) are LASSO's strongest coefficients — matching my hypothesis.

Scenario predictor

Built a what-if tool: raising Schooling 10→15 years (holding else fixed) raises predicted life expectancy by +3.5 years.

Conclusions

0.82	2.98 yrs	16 / 16	+3.5 yrs
R^2 achieved	Average error	Features kept	Schooling effect

- Schooling and Income composition of resources are the strongest forces lengthening life expectancy; Adult Mortality and HIV/AIDS are the strongest forces shortening it.
- LASSO confirmed my Linear Regression baseline was already well-specified — both models tie on every metric, with zero features needing removal.
- My results align closely with three reviewed papers, all reporting linear-model R^2 between 0.73–0.83 before ensemble methods improve further.
- Built a working “what-if” scenario tool — useful for policy questions, but I am careful to note this estimates outcomes, not future forecasts.

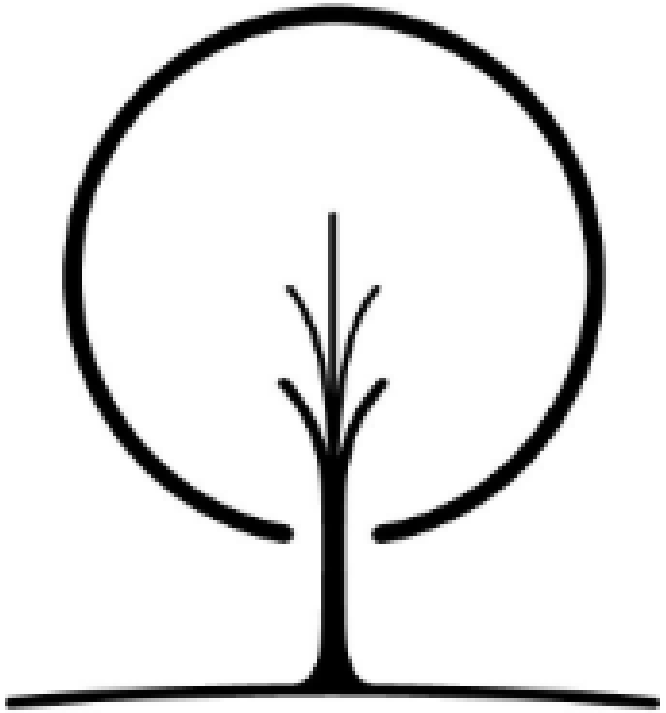


Mert Baran Demirtaş (Random Forest Regressor)

- **Performance:** Achieved a Test **R^2 score of ~0.9679** and an **RMSE of ~1.6669 (MAE: 1.0498)**.
- **Methodology:** Implemented a Random Forest Regressor optimized via *GridSearchCV*. My core contribution involved **Feature Engineering**, where I synthesized indicators such as Dev_Index and Immunity_Index to capture latent health-economic correlations.
- **Technical Outcome:** By capturing non-linear interactions through decision tree ensembles, the model significantly minimized the prediction bias found in baseline models.
- **Conclusion:** My research highlights that life expectancy is not dictated by single factors, but by the synergy between education, income, and immunization. By engineering custom indices, I was able to model these non-linear relationships with high precision. This proves that a sophisticated, data-centric approach is essential to understanding the complex fabric of global human health.

Yalın Ege Dalcan(XGBoost)

- **Performance:** Achieved a Test **R^2 score of ~0.9719** and an **RMSE of ~1.56**.
- **Methodology:** Deployed an **XGBoost (eXtreme Gradient Boosting)** regressor with a 5-Fold Cross-Validation strategy. Through extensive hyperparameter tuning (max_depth: 7, learning_rate: 0.1, n_estimators: 300), we targeted the highest possible predictive accuracy.
- **Technical Outcome:** The model proved to be the most resilient against overfitting, achieving the lowest error rate in the group by sequentially minimizing residuals and capturing even the subtle nuances of the data.
- **Conclusion:** "The XGBoost regressor provided the highest predictive power for this project, reducing the average error to just 1.56 years. This high level of accuracy validates that AI-driven gradient boosting is a powerful tool for global health forecasting. Our results suggest that we can predict long-term health trends with exceptional reliability, providing a stable foundation for future resource allocation."



General Conclusion

- Analytically, our feature correlation and model importance scores consistently highlight four critical drivers:
- **Adult Mortality (-0.70):** By far the strongest negative predictor. Our models prove that reducing adult death rates is the most efficient lever for increasing average life expectancy.
- **Schooling (+0.75):** Access to education acts as the strongest long-term foundation for health, correlating highly with better lifestyle choices and higher income potential.
- **Income Composition (+0.72):** Economic development is the engine of health, but it is not just about raw GDP—it is about how income is distributed and utilized.
- **Disease Management (HIV/AIDS -0.59):** Specific public health threats remain significant barriers, confirming that localized health crises can offset macroeconomic gains.
- **Which model is best?:** If the goal is **policy interpretation and simplicity**, Gulkhayo's **LASSO** model is the most effective tool. However, for **high-precision forecasting** and healthcare planning, Ege's **XGBoost** model is the clear winner, offering an error margin of only 1.56 years. Our final recommendation is to adopt an **ensemble-driven approach**—using the clarity of LASSO to guide policy and the predictive power of XGBoost to measure the potential impact of those policies. We have successfully translated raw WHO/UN data into a quantifiable framework for predicting, and ultimately improving, the future of human life.

